

FROGS AND TOADS



THE CROAKING SOUNDS of frogs and toads are often heard in spring as they try to attract a mate.

Frogs and toads are amphibians – cold-blooded animals that live both on land and in water. In Europe, they are easy to distinguish – frogs have slimy skins and live mainly in the water; toads have dry, warty skins and live mainly on land. In the tropics, they are more diverse and harder to tell apart. There are more than 2,600 species of frog and toad, living in most parts of the world where there is fresh water. Their habitats range from lakes and marshes, to rainforests, mountains, and deserts.

Features of frogs and toads

Frogs and toads have porous skin – water and air can pass through it – enabling them to breathe through their skin as well as their lungs. Most have sharp teeth, and can see and hear well. They have four legs, varying in length between species. Frogs range in size from a few centimetres long to the West African Goliath frog, which is 40 cm (16 in) in length. Some unusual species are the hairy frog – the male grows hair in the breeding season – and the Borneo flying frog that glides between trees.

Feeding

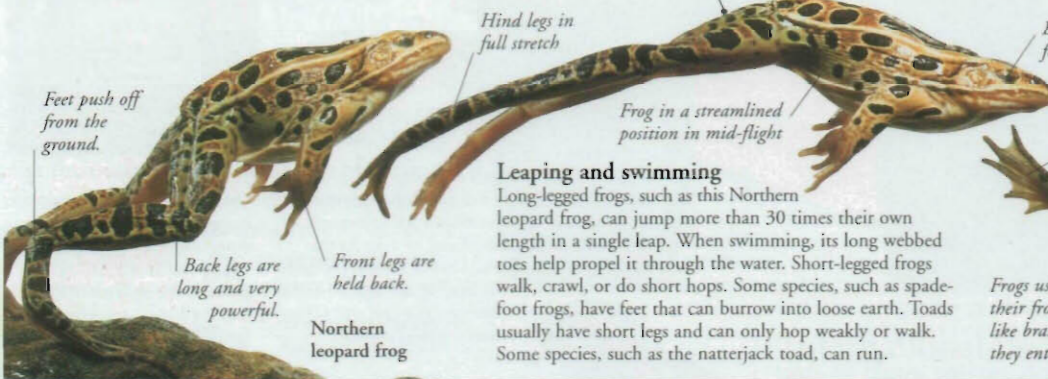
Most tadpoles are herbivorous, while adult frogs and toads are insectivorous or carnivorous. Prey includes insects, worms, spiders, fish, other frogs, or small reptiles, depending on species. Most species catch insects with their long sticky tongues. Larger frogs and toads rely on ambush. Giant horned toads and bullfrogs have powerful jaws and wide mouths and can even eat mice.



Sticky tongue is used to catch the worm.

Green toad

Short legs for hopping



Hind legs in full stretch

Frog in a streamlined position in mid-flight

Leaping and swimming

Long-legged frogs, such as this Northern leopard frog, can jump more than 30 times their own length in a single leap. When swimming, its long webbed toes help propel it through the water. Short-legged frogs walk, crawl, or do short hops. Some species, such as spadefoot frogs, have feet that can burrow into loose earth. Toads usually have short legs and can only hop weakly or walk. Some species, such as the natterjack toad, can run.

Eyes closed for protection

Webbed feet

Frogs use their front legs like brakes when they enter the water.

Frogs swim by pulling their hind legs towards their bodies, then kicking them backwards, so pushing themselves forwards through the water.

Reproduction

Most frogs and toads mate in water. The male fertilizes the eggs externally as the female lays them; a few species fertilize eggs internally. Some bear live young; others form "mating-balls" of a female and several males. The eggs develop into tadpoles that live in water and breathe through gills. The tadpoles change into air-breathing adults with lungs.



The eggs stick together.

The dark dots develop into tadpoles.

Tadpole uses its tail to swim.

Tiny buds from which front legs will grow.

Back legs

Tail eventually disappears.

Life-cycle of a common frog

1 Eggs stick together to form frogspawn. Each egg contains a dark centre that will become a tadpole. Many eggs are infertile and die, or are attacked by fungus or predators.

2 The newly hatched tadpole lives in the water and breathes through external gills. Most species at this stage are herbivorous and feed on plants.

3 By 6–9 weeks, the tadpole has grown considerably in size. The hind legs have developed. The tadpole now prepares to metamorphose, or change, into a froglet.

4 At 12 weeks, the tail has almost receded. The tadpole has become a froglet. It is now ready to leave the water to begin its adult life – partly on land.

Croaking

Frogs and toads croak to attract mates. Normally, it is the males that croak, using inflatable vocal sacs in the throat. Each species has its own distinctive croak so that the calls of the male only attract females of the same species. This avoids mating between different species in regions where there are many similar frogs and toads.

Vocal sac

Painted reed frog

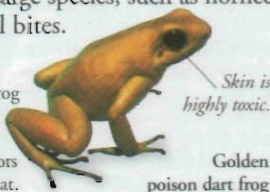


Defence

Some frogs have poisonous skin that they advertise with bright colours. Others such as tree frogs may secrete bad-tasting sticky substances. Many toads, including cane toads, have poison-secreting glands. Large species, such as horned toads, give painful bites.

Poison dart frog

This South American frog is the most poisonous frog in the world. Its bright skin warns predators that it is poisonous to eat.



Skin is highly toxic.

Golden poison dart frog

Red belly



Fire-bellied toad

Fire-bellied toads in Europe have drab backs but bright red bellies. If threatened, the toad exposes its belly. This flash of colour frightens away predators.

Tree frogs

Tree frogs move with ease within trees. They have sticky discs on their toes and an opposing thumb enabling them to grip most objects, even smooth surfaces. Flying frogs are tree frogs with very large webbed feet that enable them to glide downwards from branch to branch.



Sticky pads on their toes enable them to grip branches.

Red-eyed tree frog

NORTHERN LEOPARD FROG

SCIENTIFIC NAME *Rana pipiens*

ORDER Anura

FAMILY Ranidae

DISTRIBUTION Northern and western USA and Canada, as far south as New Mexico

HABITAT Found in most habitats, even those far from water, which explains its other name of "meadow frog"

DIET Insects

SIZE Length 9–11 cm (3.5–4.5 in)

LIFESPAN Up to 6 years (in captivity)

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